

PRF and Blockcipher

Probabilistic encryption

Encrypting the same message multiple times gives **different** ciphertexts.

$r, c \leftarrow \text{Enc}_k(r, m)$, a random r is chosen for every encryption, and sent to the receiver

$m \leftarrow \text{Dec}_k(r, c)$

Chosen-plaintext attack (CPA) security

Experiment $\text{PrivK}_{A,\Pi}^{\text{CPA}}(n)$:

1. A key k is generated using $\text{Gen}(1^n)$
2. Adversary A is given input 1^n , and **oracle access to $\text{Enc}_k(\cdot)$** . A outputs two messages $m_0, m_1 \in \mathcal{M}$ s.t. $|m_0| = |m_1|$
3. A random bit $b \in \{0,1\}$ is chosen. $c = \text{Enc}_k(m_b)$ is given to A .
4. A continues to have **oracle access to $\text{Enc}_k(\cdot)$** and outputs b'
5. Experiment outputs 1 if $b=b'$, and 0 otherwise.

Definition: a private-key encryption $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is CPA-secure, if for all probabilistic polynomial time adversaries A , there is a negligible function negl such that for all n ,

$$\Pr[\text{PrivK}_{A,\Pi}^{\text{CPA}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

or equivalently,

$$|\Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{CPA}}(c = m_0)) = 1] - \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{CPA}}(c = m_1)) = 1]| \leq \text{negl}(n)$$

CPA VS EVA

- CPA security is **strictly** stronger than EVA security

What about multiple encryptions now?

Theorem: any private-key encryption scheme that is CPA-secure, is also CPA-secure for **multiple** encryptions.

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Proof: $t = 2$, simplified hybrid proof

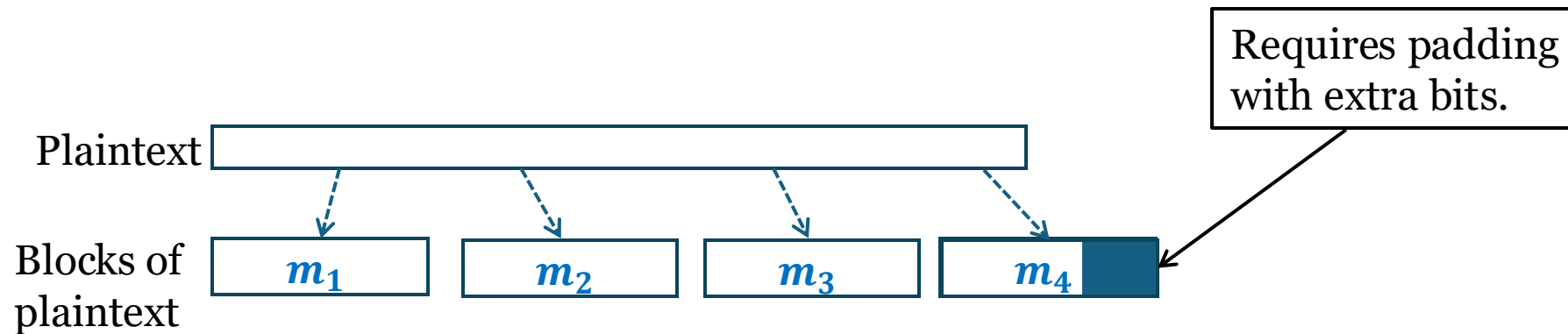
1. Adversary A is given input 1^n , and outputs two **lists** of messages $M_0 = \{m_{01}, m_{02}, \dots, m_{0t}\}$, $M_1 = \{m_{11}, m_{12}, \dots, m_{1t}\}$, s.t. $|m_{0i}| = |m_{1i}| \forall i = 1, \dots, t$
2. A key k is generated using $\text{Gen}(1^n)$, a random bit $b \in \{0,1\}$ is chosen. A list $C = \{c_1, c_2, \dots, c_t\}$ is given to A , where $c_i = \text{Enc}_k(m_{bi}) \forall i = 1, \dots, t$
3. A outputs b'
4. Experiment outputs 1 if $b=b'$, and 0 otherwise.

A private-key encryption $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is secure, if for all probabilistic polynomial time adversaries A , there is a negligible function negl such that for all n ,

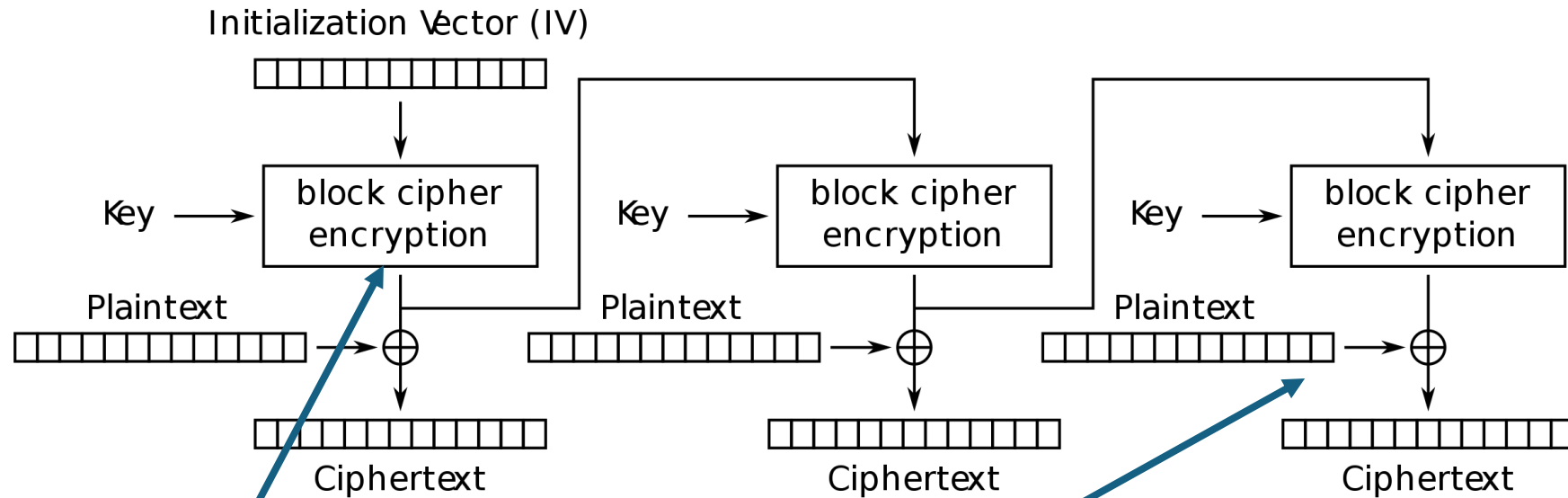
$$|\Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{CPA}}(c = M_0)) = 1] - \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{CPA}}(c = M_1)) = 1]| \leq \text{negl}(n)$$

Block Cipher

- In a **block cipher**:
 - Plaintext and ciphertext have fixed length b (e.g., 128 bits)
 - A plaintext of length n is partitioned into a sequence of m **blocks**, $P[0], \dots, P[m-1]$, where $n \leq bm < n + b$
- Each message is divided into a sequence of blocks and encrypted or decrypted in terms of its blocks.

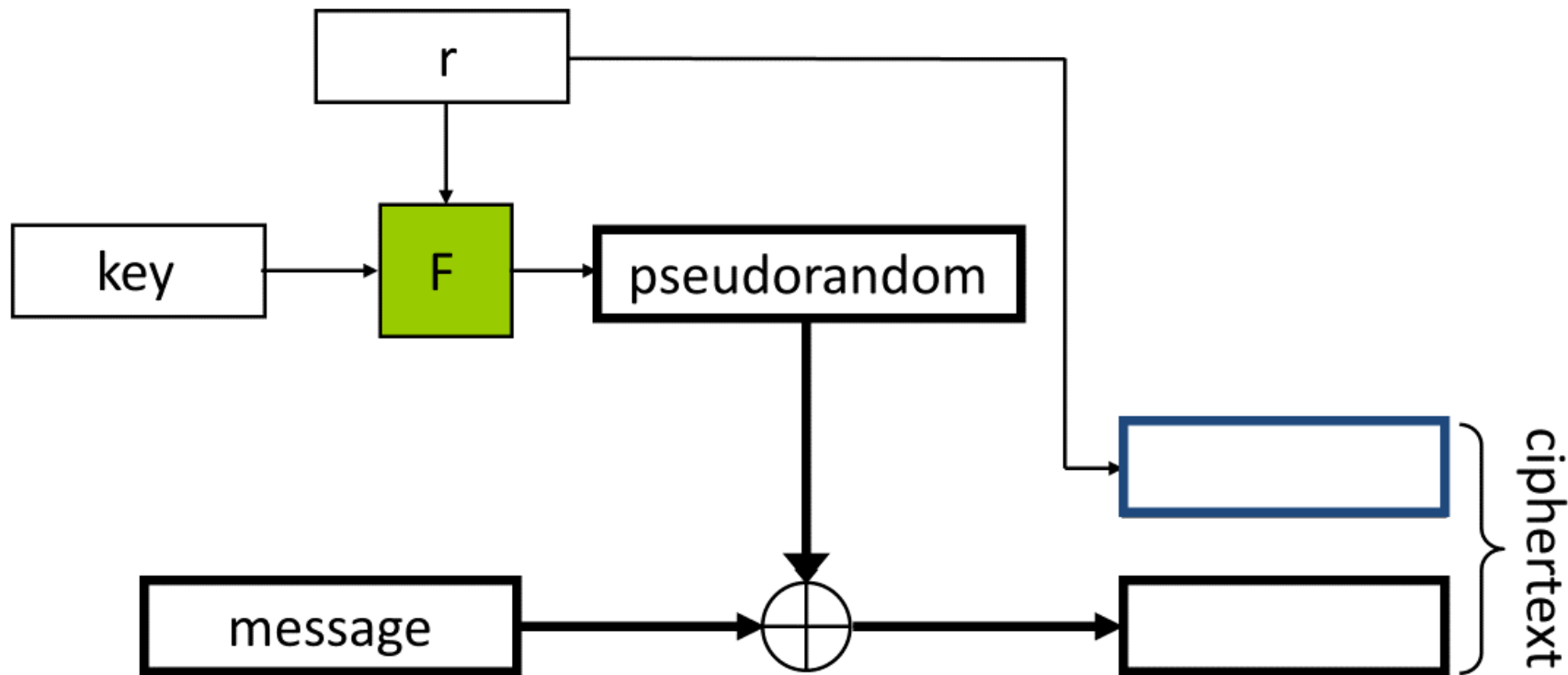


Output Feedback (OFB) mode



Output Feedback (OFB) mode encryption

1. Block cipher
2. Mode of encryption
3. Padding



$\mathbf{r}, c \leftarrow \text{Enc}_k(\mathbf{r}, m)$, a random $\mathbf{r}$ is chosen for every encryption, and sent to the receiver

$$F(k, r) = \text{PRG}(k) \oplus r \quad ?$$

Random function

- When we talk about a random function f , we mean
 1. Choosing f uniformly at random (and then fixing it)
 2. Interacting with f
- In particular, once we choose f there is no more randomness involved
 - In particular, if we query f on the same input twice, we get the same result

Number of functions

- How many functions with n bits of input and output? $F: \{0,1\}^n \rightarrow \{0,1\}^n$
- A big lookup table with 2^n entries
 1. 2^n possibilities each, so $(2^n)^{2^n} = 2^{n \cdot 2^n}$
 2. 2^n strings of size n . $n \cdot 2^n$ bits in total to represent a table. $2^{n \cdot 2^n}$ different tables.

Clicker quiz

- How many functions are there mapping $\{0,1\}^n$ to $\{0,1\}^m$?
 1. $m \cdot 2^n$
 2. $2^{n \cdot 2^m}$
 3. $m \cdot 2^{n \cdot 2^n}$
 4. $2^{m \cdot 2^n}$

Pseudorandom Function (PRF)

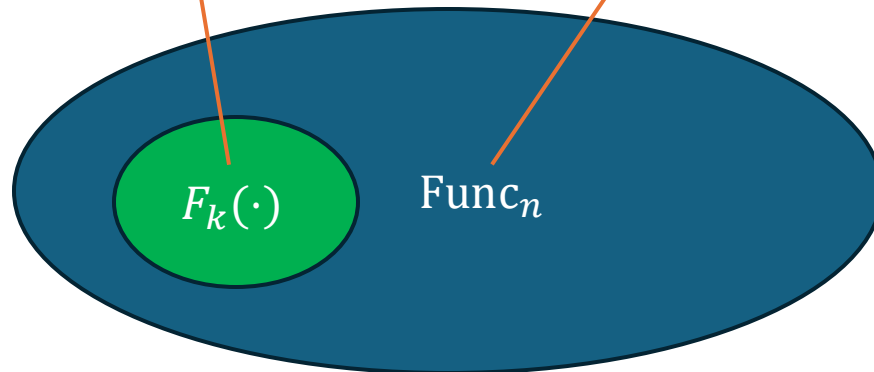
- A random function instead of a random string in PRG
- $F: \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}^n$ $F(k, x) \rightarrow y$
- Or $F_k: \{0,1\}^n \rightarrow \{0,1\}^n$ $F_k(x) \rightarrow y$
- An efficient attacker cannot differentiate it from a random function

Definition of PRF

A PRF is a function $F : \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}^n$ such that for any PPT distinguishers D , there is a negligible function negl such that

$$\left| \Pr[D^{F_k(\cdot)}(1^n) = 1] - \Pr[D^{f(\cdot)}(1^n) = 1] \right| \leq \text{negl}(n)$$

For uniformly random $k \in \{0,1\}^n$ and a random function $f \in \text{Func}_n$



PRF vs PRG

- $PRG : \{0,1\}^n \rightarrow \{0,1\}^{2n}$
 - Key space size: 2^n # of random strings: 2^{2n}
- $PRF : \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}^n$
 - Key space size: 2^n # of random functions : $2^{n \cdot 2^n}$

PRF vs PRG

PRF F immediately implies a PRG G :

Define $\text{PRG}(k) = F_k(1) \mid F_k(2) \mid F_k(3) \mid \dots$

PRF

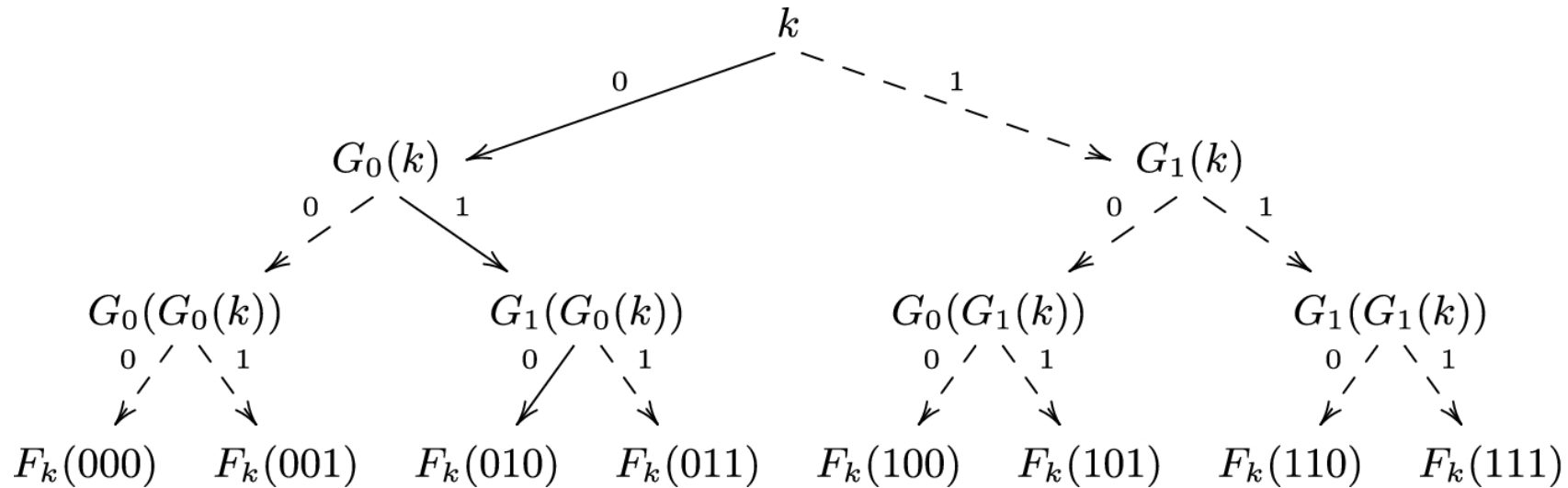


PRG



Goldreich-Goldwasser-Micali construction

Goldreich-Goldwasser-Micali construction



Lemma: Let $G(k) = G_0(k) || G_1(k)$ be a PRG from $\{0,1\}^n$ to $\{0,1\}^{2n}$. Then, for every polynomial $L = L(n)$, the following two distributions are computationally indistinguishable:

$$G(s_0), G(s_1), \dots, G(s_L) \approx r_0, r_1, \dots, r_L$$

Proof: By Hybrid argument. $G(s_0), G(s_1), \dots, G(s_L) \approx r_0, r_1, \dots, r_L$.

$$G(s_0), G(s_1), \dots, G(s_L) \approx G(s_0), G(s_1), \dots, G(s_{L-1}), r_L \approx G(s_0), G(s_1), \dots, r_{L-1}, r_L \approx, \dots, \approx r_0, r_1, \dots, r_L$$

CPA-security from PRF

Construction:

- $\text{Gen}(1^n)$: choose uniform $k \in \{0,1\}^n$ and output it as the key
- $\text{Enc}_k(m)$: on input a message $m \in \{0,1\}^n$, sample uniform $r \in \{0,1\}^n$ and compute

$$c = (r, F_k(r) \oplus m)$$

- $\text{Dec}_k(c)$: on input a ciphertext $c = (r, s)$, compute

$$m = F_k(r) \oplus s$$

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$$\Pr[\text{PrivK}_{A,\Pi}^{\text{CPA}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

Proof idea

- Reduction, build a **distinguisher** for PRF
- Build the big lookup table for function f . Give random output for the random function f , and $F_k(r)$ for PRF

Two cases: (1) r has been queried by the adversary $\rightarrow$ some gain of **success**; (2) r has not been queried by the adversary $\rightarrow$ **negl** from PRF

$$\text{Total probability: } \leq \frac{1}{2} + \frac{q(n)}{2^n} + \text{negl}(n)$$

Number of permutations

- How many one-to-one functions with n bits of input and output?

$$F: \{0,1\}^n \rightarrow \{0,1\}^n$$

- $(2^n)!$

- Stirling's approximation: $n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$

- $(2^n)! \approx \left(\frac{2^n}{e}\right)^{2^n}$

Pseudorandom Permutations

- One-to-one, a subset of PRF
- Invertible. Both $F_k(x) \rightarrow y$ and $F_k^{-1}(y) \rightarrow x$ are efficiently computable.

$$\left| \Pr \left[D^{F_k(\cdot), F_k^{-1}(\cdot)}(1^n) = 1 \right] - \Pr \left[D^{f(\cdot), f^{-1}(\cdot)}(1^n) = 1 \right] \right| \leq \text{negl}(n)$$